

# Calculus for Biology and Medicine, 3<sup>rd</sup> Edition, Claudia Neuhauser

11.2 #1, 3, 6, 9, 12

1.  $A = \begin{bmatrix} -(a+c) & b \\ a & -(b+d) \end{bmatrix}$

$$\frac{d\vec{x}}{dt} = \begin{bmatrix} -(1.7+0.1) & 0.6 \\ 1.7 & -(0.6+0.3) \end{bmatrix} \vec{x}$$

$$\frac{d\vec{x}}{dt} = \begin{bmatrix} -1.8 & 0.6 \\ 1.7 & -0.9 \end{bmatrix} \vec{x}$$

$$\Delta = ad + bc + cd$$

$$= (0.5)(0.02) + (0.1)(0.05) + (0.05)(0.02) > 0 \Rightarrow \text{stable node}$$

3.  $A = \begin{bmatrix} -(a+c) & b \\ a & -(b+d) \end{bmatrix}$

$$\frac{d\vec{x}}{dt} = \begin{bmatrix} -(2.5+0) & 0.7 \\ 2.5 & -(0.7+0.1) \end{bmatrix} \vec{x}$$

$$\frac{d\vec{x}}{dt} = \begin{bmatrix} -2.5 & 0.7 \\ 2.5 & -0.8 \end{bmatrix} \vec{x}$$

$$\Delta = ad + bc + cd$$

$$= (2.5)(0.1) + (0.7)(0) + (0)(0.1) > 0 \Rightarrow \text{stable node}$$

6.  $A = \begin{bmatrix} -(a+c) & b \\ a & -(b+d) \end{bmatrix}$

$$\frac{d\vec{x}}{dt} = \begin{bmatrix} -(0.2+0) & 0.1 \\ 0.2 & -(0.1+0) \end{bmatrix} \vec{x}$$

$$\frac{d\vec{x}}{dt} = \begin{bmatrix} -0.2 & 0.1 \\ 0.2 & -0.1 \end{bmatrix} \vec{x}$$

$$\Delta = ad + bc + cd$$

$$= (0.2)(0) + (0.1)(0) + (0)(0) = 0 \Rightarrow \text{conserved}$$

9.  $\frac{dx_1}{dt} = -0.4x_1 + 0.3x_2$   $\frac{dx_2}{dt} = 0.1x_1 - 0.5x_2$

Diagram showing transitions between  $x_1$  and  $x_2$ :

- $x_1 \xrightarrow{a=0.1x_1/\text{hr}} x_2$
- $x_2 \xrightarrow{b=0.3x_2/\text{hr}} x_1$
- $x_1 \xrightarrow{c=0.3x_1/\text{hr}}$  (out)
- $x_2 \xrightarrow{d=0.2x_2/\text{hr}}$  (out)

$-(a+c) = -0.4$     $-(b+d) = -0.5$

$a = 0.1$     $c = 0.3$     $b = 0.3$     $d = 0.2$

12.  $\frac{dx_1}{dt} = -0.2x_1 + 1.1x_2$   $\frac{dx_2}{dt} = 0.2x_1 - 1.1x_2$

Diagram showing transitions between  $x_1$  and  $x_2$ :

- $x_1 \xrightarrow{a=0.2x_1/\text{hr}} x_2$
- $x_2 \xrightarrow{b=1.1x_2/\text{hr}} x_1$
- $x_1 \xrightarrow{c=0x_1/\text{hr}}$  (out)
- $x_2 \xrightarrow{d=0x_2/\text{hr}}$  (out)

$-(a+c) = -0.2$     $-(b+d) = -1.1$

$a = 0.2$     $c = 0$     $b = 1.1$     $d = 0$